PAGE	NAME
01	Title Block
02	Block Diagram
03	Power In
04	FPGA Power
05	FPGA GPIOs1
06	FPGA GPIOs2
07	DDR3
08	microSD
09	HDMI
10	UART
11	JTAG
12	Pinout

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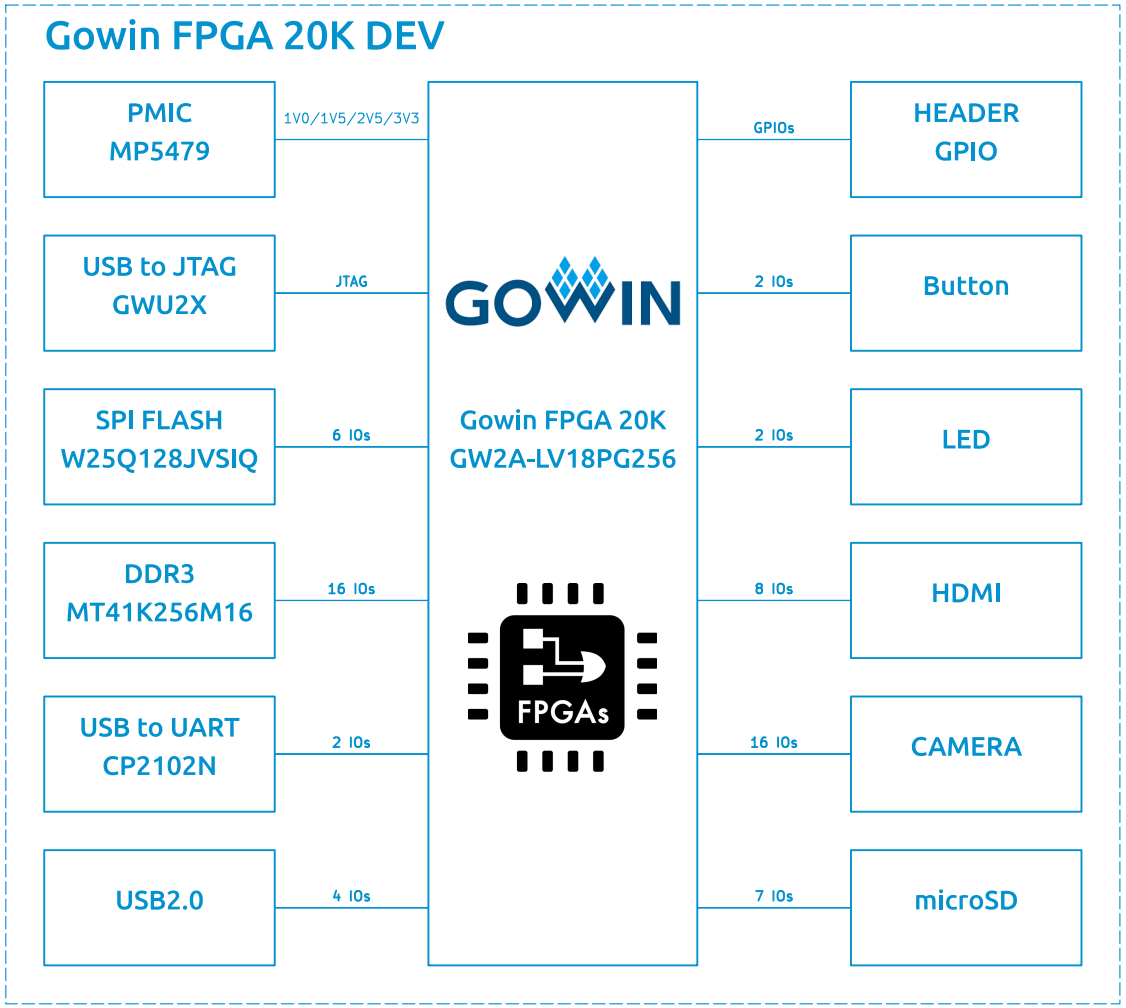
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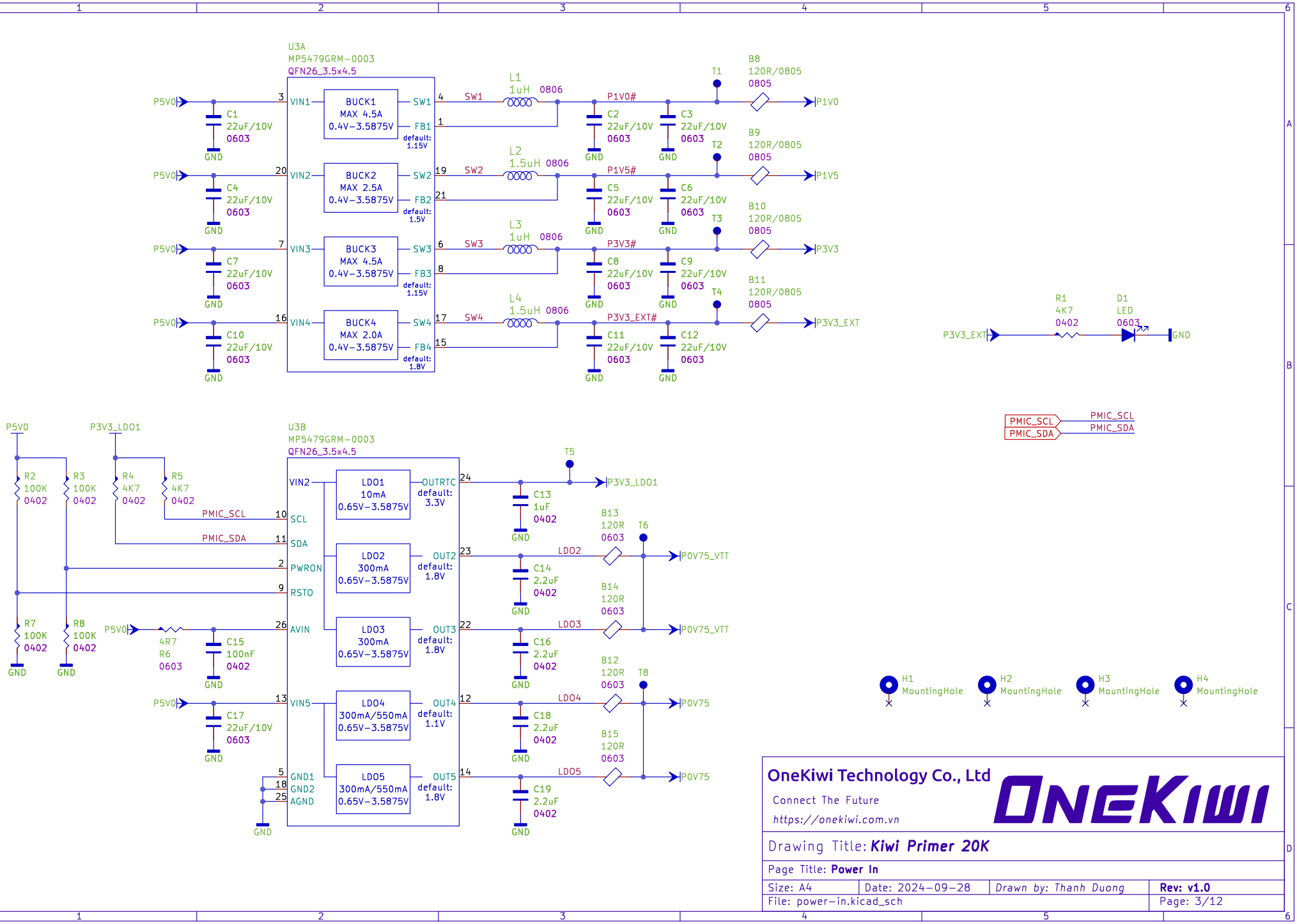
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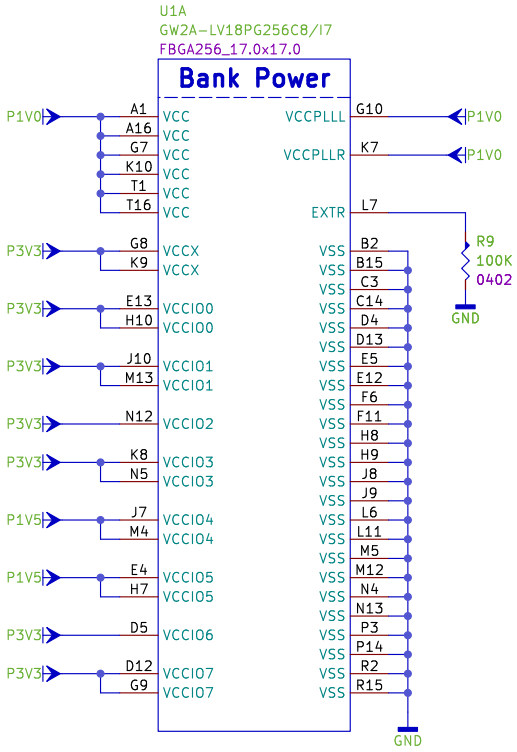
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Size: A4	Date: 2024-09-28	Drawn by: Thanh Duong	**Rev: v1.0**
File: fpga-kiwi-primer-20k.kicad_sch			Page: 1/12

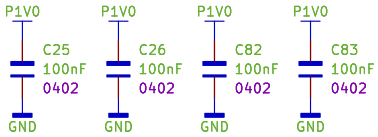


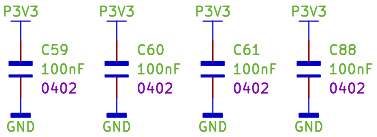
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Size: A4	Date: 2024-09-28	*Drawn by: Thanh Duong*	**Rev: v1.0**
File: block-diagram.kicad_sch			Page: 2/12

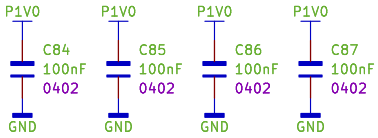

U3A
MP5479GRM-0003
QFN26_3.5x4.5
BUCK1
MAX 4.5A
0.4V-3.5875V
BUCK2
MAX 2.5A
0.4V-3.5875V
BUCK3
MAX 4.5A
0.4V-3.5875V
BUCK4
MAX 2.0A
0.4V-3.5875V
VIN1
VIN2
VIN3
VIN4
SW1
SW2
SW3
SW4
FB1
FB2
FB3
FB4
default: 1.15V
default: 1.5V
default: 1.15V
default: 1.8V
P5V0
L1 1uH 0806
L2 1.5uH 0806
L3 1uH 0806
L4 1.5uH 0806
P1V0#
P1V5#
P3V3#
P3V3_EXT#
C1 22uF/10V 0603
C2 22uF/10V 0603
C3 22uF/10V 0603
C4 22uF/10V 0603
C5 22uF/10V 0603
C6 22uF/10V 0603
C7 22uF/10V 0603
C8 22uF/10V 0603
C9 22uF/10V 0603
C10 22uF/10V 0603
C11 22uF/10V 0603
C12 22uF/10V 0603
T1
T2
T3
T4
B8 120R/0805 0805
B9 120R/0805 0805
B10 120R/0805 0805
B11 120R/0805 0805
P1V0
P1V5
P3V3
P3V3_EXT
GND
R1 4K7 0402
D1 LED 0603
P3V3_EXT
GND
PMIC_SCL
PMIC_SDA
U3B
MP5479GRM-0003
QFN26_3.5x4.5
P5V0
P3V3_LDO1
R2 100K 0402
R3 100K 0402
R4 4K7 0402
R5 4K7 0402
R7 100K 0402
R8 100K 0402
R6 4R7 0603
C15 100nF 0402
C17 22uF/10V 0603
LDO1 10mA 0.65V-3.5875V
LDO2 300mA 0.65V-3.5875V
LDO3 300mA 0.65V-3.5875V
LDO4 300mA/550mA 0.65V-3.5875V
LDO5 300mA/550mA 0.65V-3.5875V
VIN2
VIN5
SCL
SDA
PWRON
RSTO
AVIN
GND1
GND2
AGND
OUTRTC default: 3.3V
OUT2 default: 1.8V
OUT3 default: 1.8V
OUT4 default: 1.1V
OUT5 default: 1.8V
C13 1uF 0402
C14 2.2uF 0402
C16 2.2uF 0402
C18 2.2uF 0402
C19 2.2uF 0402
T5
T6
T8
B13 120R 0603
B14 120R 0603
B12 120R 0603
B15 120R 0603
LDO2
LDO3
LDO4
LDO5
P0V75_VTT
P0V75
H1 MountingHole
H2 MountingHole
H3 MountingHole
H4 MountingHole
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Drawing Title: Kiwi Primer 20K
Page Title: Power In
Size: A4
Date: 2024-09-28
Drawn by: Thanh Duong
Rev: v1.0
File: power-in.kicad_sch
Page: 3/12

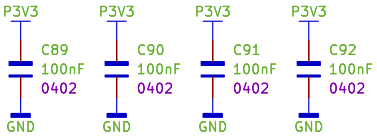


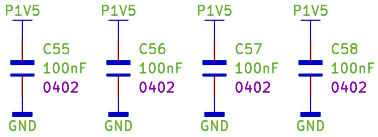
Power GW2A-18 Package PG256			
Name	**Description**	**Min**	**Max**
VCC	Core voltage	0.95V	1.05V
VCCIO0~7	I/O Bank voltage	1.14V	3.6V
VCCX	Auxiliary voltage	2.7V	3.6V
VCCPLLL	Left PLL supply voltage	0.95V	1.05V
VCCPLLR	Right PLL supply voltage	0.95V	1.05V











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Drawing Title: **Kiwi Primer 20K**

Page Title: **FPGA Power**

Size: A4	Date: 2024-09-28	Drawn by: Thanh Duong	**Rev: v1.0**
File: fpga-power.kicad_sch			Page: 4/12

U1B
GW2A-LV18PG256C8/I7
FBGA256_17.0x17.0

Bank 0

BO_T2_P	L15	IOT2A/DQ15/LVDS
HDMI_TX2_P	D16	IOT4A/DQ15/LVDS
HDMI_TX2_N	E14	IOT4B/DQ15/LVDS
LED2	C16	IOT5A/DQ15
LED1	D15	IOT5B/DQ15
HDMI_TX1_P	E16	IOT6A/DQ15/LVDS
HDMI_TX1_N	F15	IOT6B/DQ15/LVDS
LED8	F13	IOT8A/DQ15/LVDS
BO_T8_N	G12	IOT8B/DQ15/LVDS
	F14	IOT9A/DQS15
HDMI_CEC	F16	IOT9B/DQS15
BO_T12_P	F12	IOT12A/DQ14/LVDS
LED7	G13	IOT12B/DQ14/LVDS
LED5	G15	IOT13A/DQ14
LED6	G14	IOT13B/DQ14
BO_T14_P	G11	IOT14A/DQ14/LVDS
BO_T14_N	H12	IOT14B/DQ14/LVDS
HDMI_TX0_P	G16	IOT16A/DQ14/LVDS
HDMI_TX0_N	H15	IOT16B/DQ14/LVDS
LED4	H13	IOT18A/DQ14/LVDS
BO_T18_N	J12	IOT18B/DQ14/LVDS
LED3	H14	IOT20A/DQ14/LVDS
HDMI_SCL	H16	IOT20B/DQ14/LVDS
HDMI_HPD	J16	IOT22A/DQ14/LVDS
HDMI_SDA	J14	IOT22B/DQ14/LVDS
HDMI_TXC_P	J15	IOT24A/DQ14/LVDS
HDMI_TXC_N	K16	IOT24B/DQ14/LVDS
SYS_CLK	H11	IOT27A/DQ14/GCLKT_0
BO_T27_N	J13	IOT27B/DQ14/GCLKC_0

U1C
GW2A-LV18PG256C8/I7
FBGA256_17.0x17.0

Bank 1

B1_T30_P	K14	IOT30A/DQ13/GCLKT_1/LVDS
B1_T30_N	K15	IOT30B/DQ13/GCLKC_1/LVDS
B1_T32_P	J11	IOT32A/DQ13/LVDS
B1_T32_N	L12	IOT32B/DQ13/LVDS
B1_T34_P	L16	IOT34A/DQ13/LVDS
B1_T34_N	L14	IOT34B/DQ13/LVDS
B1_T36_P	K13	IOT36A/DQ13/LVDS
B1_T36_N	K12	IOT36B/DQ13/LVDS
B1_T38_P	K11	IOT38A/DQ13/LVDS
B1_T38_N	L13	IOT38B/DQ13/LVDS
B1_T40_P	M14	IOT40A/DQ13/LVDS
B1_T40_N	M15	IOT40B/DQ13/LVDS
B1_T44_P	D14	IOT44A/DQ13/LVDS
B1_T44_N	E15	IOT44B/DQ13/LVDS
B1_T48_P	N15	IOT48A/DQS12/LVDS
B1_T48_N	P16	IOT48B/DQS12/LVDS
B1_T52_P	N16	IOT52A/DQ12/LVDS
B1_T52_N	N14	IOT52B/DQ12/LVDS
B1_T54_P	P15	IOT54A/DQ12/LVDS
B1_T54_N	R16	IOT54B/DQ12/LVDS

U1D
GW2A-LV18PG256C8/I7
FBGA256_17.0x17.0

Bank 2

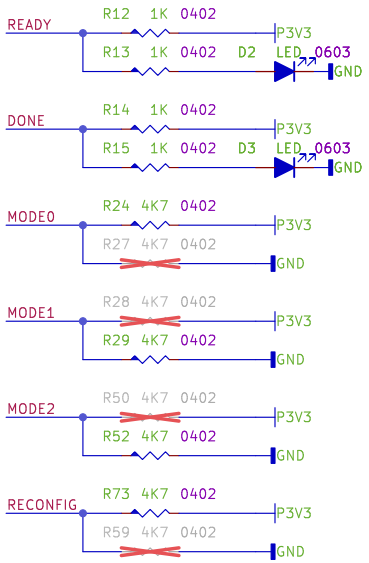
B2_R7_P	T15	IOR7A/DQ11/RPLL1_T_in
B2_R7_N	R14	IOR7B/DQ11/RPLL1_C_in
B2_R8_P	P12	IOR8A/DQ11/RPLL1_T_fb/LVDS
B2_R8_N	T13	IOR8B/DQ11/RPLL1_C_fb/LVDS
B2_R11_P	R12	IOR11A/DQ10/LVDS
B2_R11_N	P13	IOR11B/DQ10/LVDS
B2_R17_P	R11	IOR17A/DQ10/LVDS
B2_R17_N	T12	IOR17B/DQ10/LVDS
B2_R20_P	R13	IOR20A/DQ10/LVDS
B2_R20_N	T14	IOR20B/DQ10/LVDS
B2_R22_P	M10	IOR22A/DQS10/LVDS
B2_R22_N	N11	IOR22B/DQS10/LVDS
B2_R24_P	T11	IOR24A/DQ10/LVDS
B2_R24_N	P11	IOR24B/DQ10/LVDS
FPGA_TDO	C6	IOR25A/DQ10/TDO
FPGA_TMS	B8	IOR25B/DQ10/TMS
FPGA_TCK	A7	IOR26A/DQ10/TCK/LVDS
FPGA_TDI	A6	IOR26B/DQ10/TDI/LVDS
B2_R27_P	N10	IOR27A/DQ10/GCLKT_2
B2_R27_N	M11	IOR27B/DQ10/GCLKC_2

U1E
GW2A-LV18PG256C8/I7
FBGA256_17.0x17.0

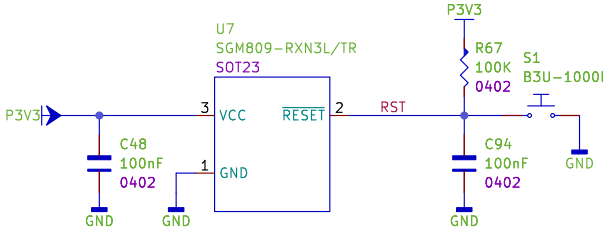
Bank 3

USB1_P	T7	IOR29A/DQ9/GCLKT_3/LVDS
USB1_N	R8	IOR29B/DQ9/GCLKC_3/LVDS
MODE0	M16	IOR30A/MODE0/DQ9
MODE1	B16	IOR30B/MODE1/DQ9
MODE2	C15	IOR31A/MODE2/DQ9/LVDS
RECONFIG	B10	IOR31B/RECONFIG_N/DQ9/LVDS
READY	A13	IOR32A/DQ9/READY
DONE	C13	IOR32B/DQ9/DONE
MSPI_MISO	P10	IOR33A/DQ9/MI/D7/LVDS
MSPI_MOSI	R10	IOR33B/DQ9/MO/D6/LVDS
MSPI_CS	M9	IOR34A/DQ9/MCS_N/D5
MSPI_MCLK	L10	IOR34B/DQ9/MCLK/D4
B3_R35_P	R9	IOR35A/DQ9/FASTRD_N/D3/LVDS
B3_R35_N	T10	IOR35B/DQ9/SI/D2/LVDS
B3_R36_P	M8	IOR36A/DQ9/SO/D1
B3_R36_N	N9	IOR36B/DQ9/SSPI_CS_N/D0
B3_R38_P	T9	IOR38A/DQ9/DIN/CLKHOLD_N/LVDS
B3_R38_N	P9	IOR38B/DQ9/DOUT/WE_N/LVDS
B3_R39_P	C10	IOR39A/DQ9/SCLK
B3_R40_P	N8	IOR40A/DQ9/LVDS
B3_R40_N	L9	IOR40B/DQ9/LVDS
LCD_SDA	P8	IOR42A/DQ9/LVDS
LCD_SCL	T8	IOR42B/DQ9/LVDS
B3_R44_P	M6	IOR44A/DQ9/LVDS
B3_R44_N	L8	IOR44B/DQ9/LVDS
B3_R47_P	M7	IOR47A/DQ8/RPLL2_T_fb/LVDS
B3_R47_N	N7	IOR47B/DQ8/RPLL2_C_fb/LVDS
B3_R49_P	R7	IOR49A/DQ8/LVDS
B3_R49_N	P7	IOR49B/DQ8/LVDS
B3_R51_P	N6	IOR51A/DQ8/LVDS
USB0_P	P6	IOR53A/DQ8/LVDS
USB0_N	T6	IOR53B/DQ8/LVDS

CONFIG

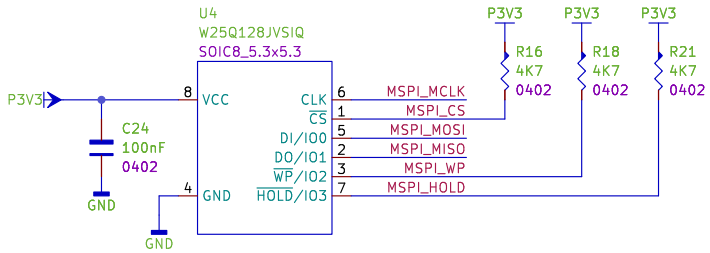


U7
SGM809-RXN3L/TR
SOT23

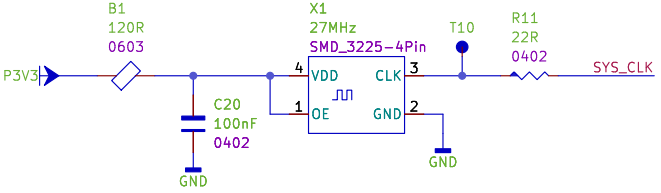


FLASH

U4
W25Q128JVSIQ
SOIC8_5.3x5.3



CLOCK 27MHz



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Drawing Title: ***Kiwi Primer 20K***

Page Title: **FPGA IOs1**

Size: A4	Date: 2024-09-28	*Drawn by: Thanh Duong*	**Rev: v1.0**
File: fpga-ios1.kicad_sch			Page: 5/12

DQ[7:0],LDQSn,LDQSp and LDM is assigned to Bank5's DQ5
DQ[15:8],UDQSn,UDQSp and UDM is assigned to Bank5's DQ4

U1F
GW2A-LV18PG256C8/I7
FBGA256_17.0x17.0

Bank 4

DDR_BA0	L2	IOB30A/DQ6/GCLKT_4/LVDS
DDR_BA2	M1	IOB30B/DQ6/GCLKC_4/LVDS
	H3	IOB32A/DQ6/LVDS
DDR_CKE	H1	IOB32B/DQ6/LVDS
	J2	IOB34A/DQ6/LVDS
DDR_BA1	K1	IOB34B/DQ6/LVDS
	H5	IOB35A/DQ6
DDR_ODT	J4	IOB35B/DQ6
DDR_CAS	K3	IOB36A/DQ6/LVDS
DDR_RAS	K2	IOB36B/DQ6/LVDS
	J5	IOB37A/DQS6
	K6	IOB37B/DQS6
DDR_WE	L1	IOB38A/DQ6/LVDS
DDR_A10	L3	IOB38B/DQ6/LVDS
DDR_CS	K4	IOB39A/DQ6
	L5	IOB39B/DQ6
	K5	IOB40A/DQ6/LVDS
	L4	IOB40B/DQ6/LVDS
DDR_A9	N2	IOB41A/DQ6
DDR_A7	P1	IOB41B/DQ6
DDR_A6	M3	IOB42A/DQ6/LVDS
DDR_A4	N1	IOB42B/DQ6/LVDS
DDR_A12	M2	IOB43A/DQ6
DDR_A1	N3	IOB43B/DQ6
DDR_RST	R1	IOB44A/DQ6/LVDS
DDR_A13	P2	IOB44B/DQ6/LVDS
DDR_A11	P4	IOB45A/DQ6
DDR_A3	T4	IOB45B/DQ6
DDR_A0	R3	IOB48A/DQS7/LVDS
DDR_A2	T2	IOB48B/DQS7/LVDS
	P5	IOB50A/DQ7/LVDS
	R5	IOB50B/DQ7/LVDS
DDR_A8	R4	IOB52A/DQ7/LVDS
DDR_A5	T3	IOB52B/DQ7/LVDS
	R6	IOB54A/DQ7/LVDS
	T5	IOB54B/DQ7/LVDS

U1G
GW2A-LV18PG256C8/I7
FBGA256_17.0x17.0

Bank 5

DDR_DQ15	A4	IOB2A/DQ4/LVDS
DDR_DQ14	C5	IOB2B/DQ4/LVDS
DDR_DQ8	D6	IOB3A/DQ4
	E7	IOB3B/DQ4
DDR_DQ9	A3	IOB4A/DQ4/LVDS
DDR_DQ12	B4	IOB4B/DQ4/LVDS
DDR_DQ13	A5	IOB7A/DQ4
DDR_DQ11	B6	IOB7B/DQ4
DDR_DQM1	B1	IOB8A/DQ4/LVDS
DDR_DQ10	C2	IOB8B/DQ4/LVDS
DDR_DQS1_P	D3	IOB9A/DQS4
DDR_DQS1_N	D1	IOB9B/DQS4
DDR_DQM0	E2	IOB12A/DQ5/LVDS
	E3	IOB12B/DQ5/LVDS
	B3	IOB13A/DQ5
	A2	IOB13B/DQ5
	C1	IOB14A/DQ5/LVDS
	D2	IOB14B/DQ5/LVDS
DDR_DQ2	E1	IOB16A/DQ5/LVDS
DDR_DQ0	F2	IOB16B/DQ5/LVDS
	F4	IOB18A/DQ5/LVDS
	G6	IOB18B/DQ5/LVDS
DDR_DQ5	F3	IOB19A/DQ5
DDR_DQ1	F1	IOB19B/DQ5
	G5	IOB20A/DQ5/LVDS
DDR_DQ7	G4	IOB20B/DQ5/LVDS
DDR_DQS0_P	G2	IOB21A/DQS5
DDR_DQS0_N	G3	IOB21B/DQS5
	F5	IOB22A/DQ5/LVDS
	H6	IOB22B/DQ5/LVDS
DDR_DQ3	G1	IOB24A/DQ5/LVDS
DDR_DQ4	H2	IOB24B/DQ5/LVDS
DDR_DQ6	H4	IOB26A/DQ5/LVDS
	J6	IOB26B/DQ5/LVDS
DDR_CLK_P	J1	IOB27A/DQ5/GCLKT_5
DDR_CLK_N	J3	IOB27B/DQ5/GCLKC_5

U1H
GW2A-LV18PG256C8/I7
FBGA256_17.0x17.0

Bank 6

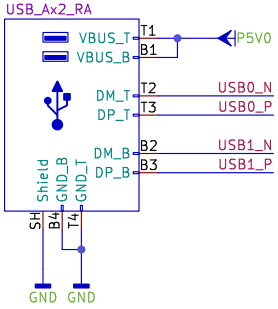
B6_L29_P	C8	IOL29A/DQ2/GCLKT_6/LVDS
B6_L29_N	A8	IOL29B/DQ2/GCLKC_6/LVDS
B6_L31_P	F9	IOL31A/DQ2/LVDS
B6_L31_N	E11	IOL31B/DQ2/LVDS
B6_L33_P	B9	IOL33A/DQ2/LVDS
B6_L33_N	A10	IOL33B/DQ2/LVDS
B6_L35_P	F8	IOL35A/DQ2/LVDS
B6_L35_N	D9	IOL35B/DQ2/LVDS
B6_L38_P	D8	IOL38A/DQ2/LVDS
B6_L38_N	E9	IOL38B/DQ2/LVDS
B6_L40_P	B7	IOL40A/DQ2/LVDS
B6_L40_N	C7	IOL40B/DQ2/LVDS
B6_L45_P	F7	IOL45A/DQ2/LPLL2_T_in
B6_L45_N	E8	IOL45B/DQ2/LPLL2_C_in
UART_RX	C4	IOL47A/DQ3/LPLL2_T_fb/LVDS
UART_TX	B5	IOL47B/DQ3/LPLL2_C_fb/LVDS
B6_L53_P	E6	IOL53A/DQ3/LVDS
B6_L53_N	D7	IOL53B/DQ3/LVDS

U1I
GW2A-LV18PG256C8/I7
FBGA256_17.0x17.0

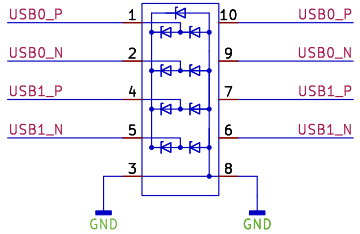
Bank 7

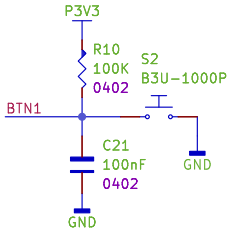
SDIO_D0	B14	IOL2A/DQ0/LVDS
SDIO_DET	A15	IOL2B/DQ0/LVDS
SDIO_CMD	C12	IOL7A/DQ0/LPLL1_T_in
BTN2	B12	IOL7B/DQ0/LPLL1_C_in
WS2812_IN	B13	IOL8A/DQ0/LPLL1_T_fb/LVDS
SDIO_D1	A14	IOL8B/DQ0/LPLL1_C_fb/LVDS
B7_L11_P	F10	IOL11A/DQ1/LVDS
RST	B11	IOL13A/DQ1/LVDS
BTN1	A12	IOL13B/DQ1/LVDS
BTN3	A11	IOL15A/DQ1/LVDS
SDIO_D2	C11	IOL15B/DQ1/LVDS
B7_L17_P	D10	IOL17A/DQ1/LVDS
B7_L17_N	E10	IOL17B/DQ1/LVDS
SDIO_D3	D11	IOL22A/DQS1/LVDS
SDIO_CLK	A9	IOL27A/DQ1/GCLKT_7
B7_L27_N	C9	IOL27B/DQ1/GCLKC_7

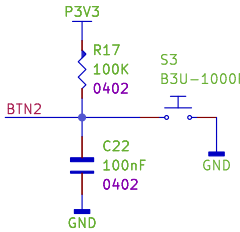
P1
USB_Ax2
USB_Ax2_RA

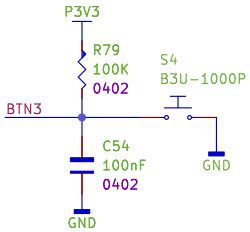


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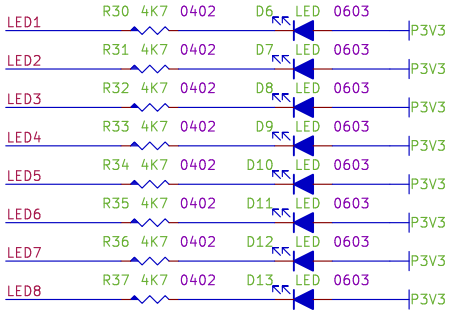








LED



USB Host





USB Device



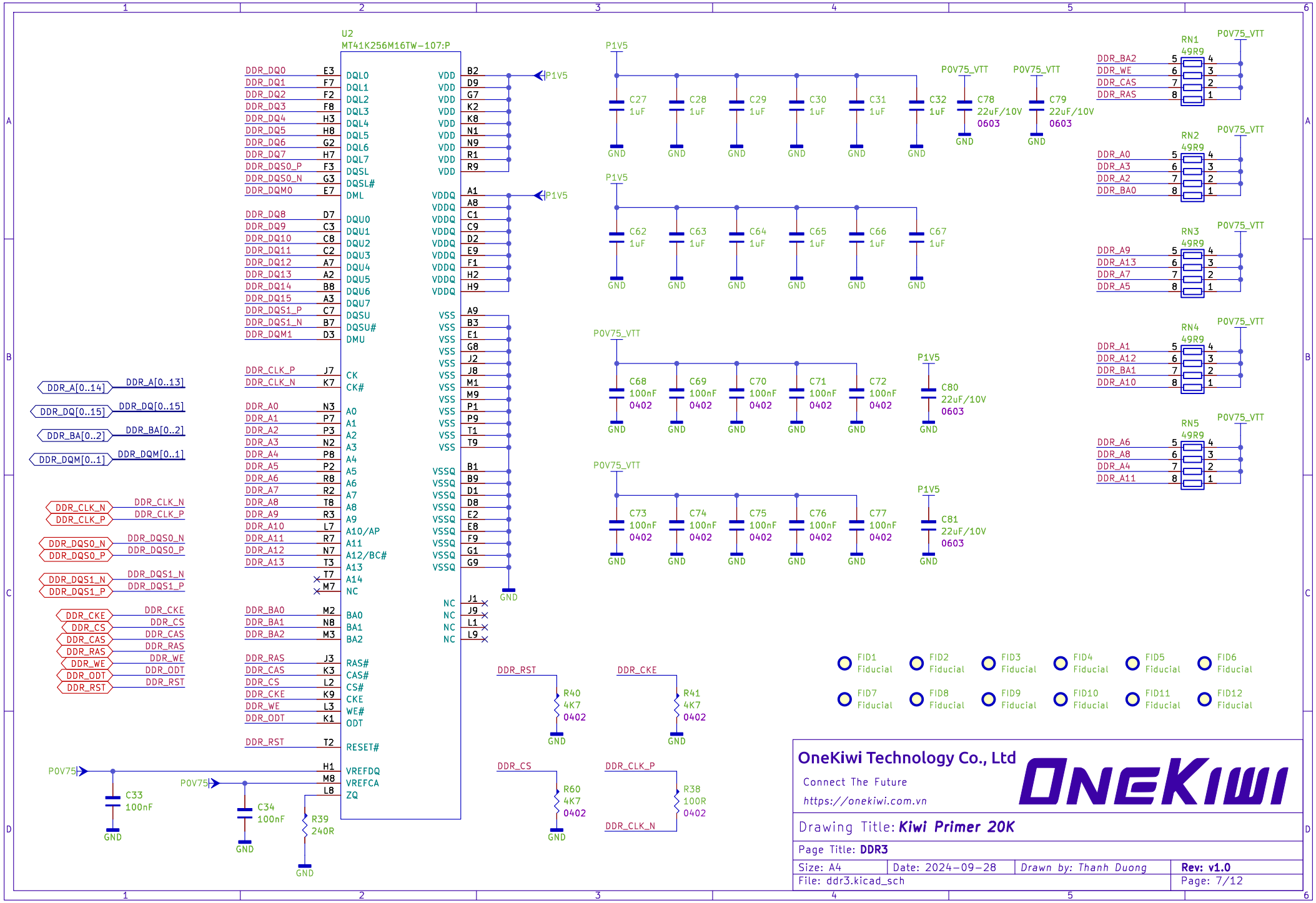


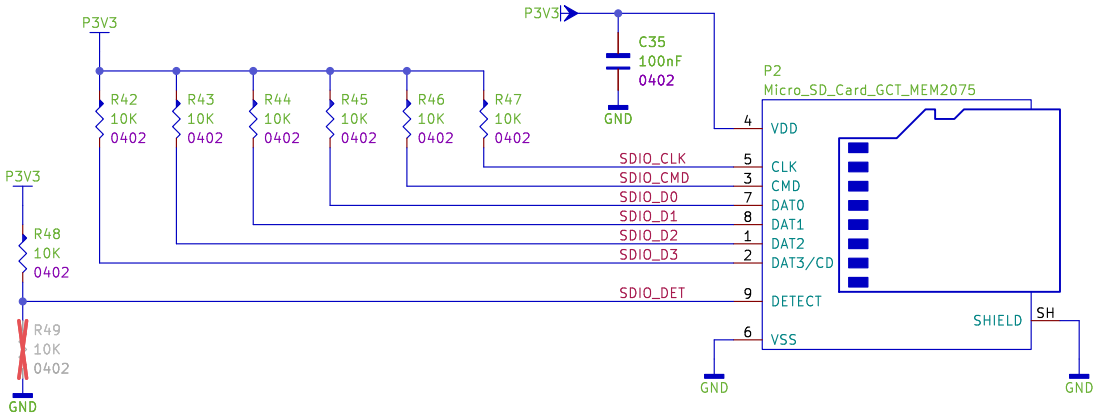
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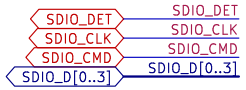
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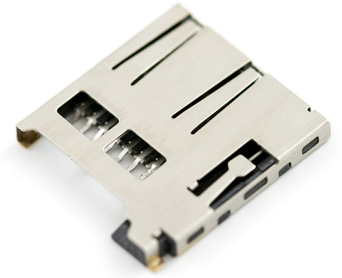
Page Title: **FPGA IOs2**

Size: A4	Date: 2024-09-28	*Drawn by: Thanh Duong*	**Rev: v1.0**
File: fpga-ios2.kicad_sch			Page: 6/12


U2
MT41K256M16TW-107:P
DDR_DQ0 DDR_DQ1 DDR_DQ2 DDR_DQ3 DDR_DQ4 DDR_DQ5 DDR_DQ6 DDR_DQ7
DDR_DQS0_P DDR_DQS0_N DDR_DQM0
DDR_DQ8 DDR_DQ9 DDR_DQ10 DDR_DQ11 DDR_DQ12 DDR_DQ13 DDR_DQ14 DDR_DQ15
DDR_DQS1_P DDR_DQS1_N DDR_DQM1
DDR_CLK_P DDR_CLK_N
DDR_A0 DDR_A1 DDR_A2 DDR_A3 DDR_A4 DDR_A5 DDR_A6 DDR_A7 DDR_A8 DDR_A9 DDR_A10 DDR_A11 DDR_A12 DDR_A13
DDR_BA0 DDR_BA1 DDR_BA2
DDR_RAS DDR_CAS DDR_CS DDR_CKE DDR_WE DDR_ODT
DDR_RST
P0V75
P1V5
GND
DDR_A[0..14] DDR_A[0..13]
DDR_DQ[0..15] DDR_DQ[0..15]
DDR_BA[0..2] DDR_BA[0..2]
DDR_DQM[0..1] DDR_DQM[0..1]
C27 C28 C29 C30 C31 C32 1uF
C78 C79 22uF/10V 0603
C62 C63 C64 C65 C66 C67 1uF
C68 C69 C70 C71 C72 100nF 0402
C80 22uF/10V 0603
C73 C74 C75 C76 C77 100nF 0402
C81 22uF/10V 0603
C33 C34 100nF
R39 240R
R40 R41 4K7 0402
R60 4K7 0402
R38 100R 0402
RN1 RN2 RN3 RN4 RN5 49R9
FID1 FID2 FID3 FID4 FID5 FID6 FID7 FID8 FID9 FID10 FID11 FID12 Fiducial
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Drawing Title: Kiwi Primer 20K
Page Title: DDR3
Size: A4
Date: 2024-09-28
Drawn by: Thanh Duong
Rev: v1.0
File: ddr3.kicad_sch
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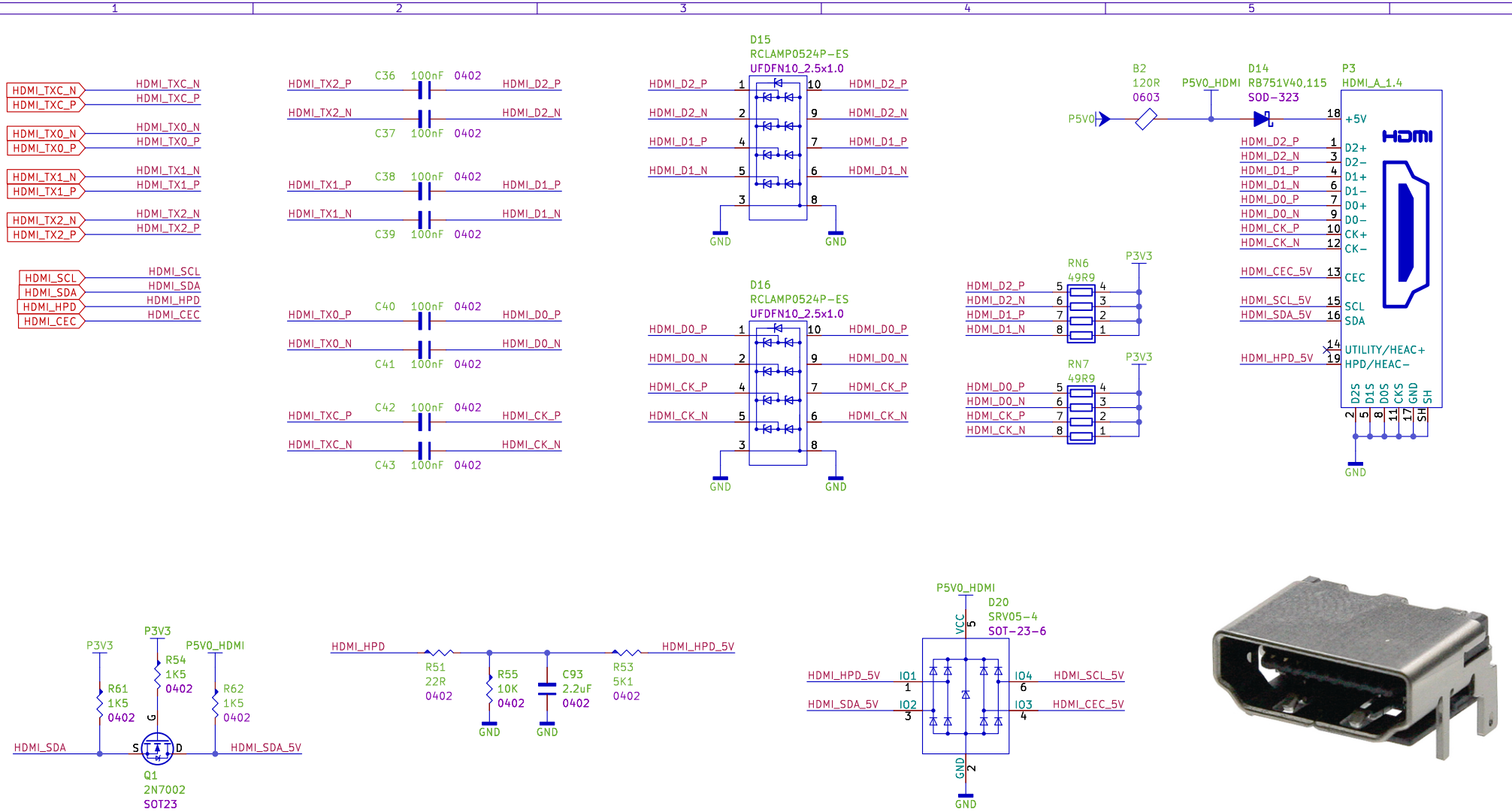
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Drawing Title: ***Kiwi Primer 20K***

Page Title: **microSD**

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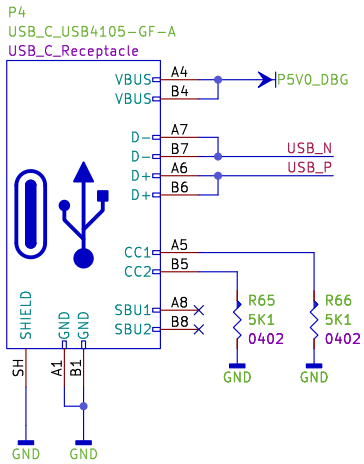
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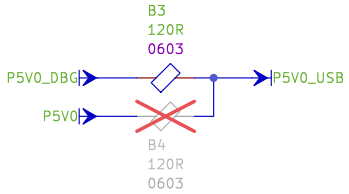
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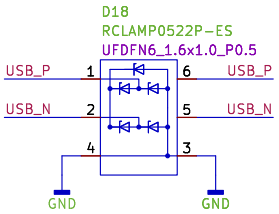
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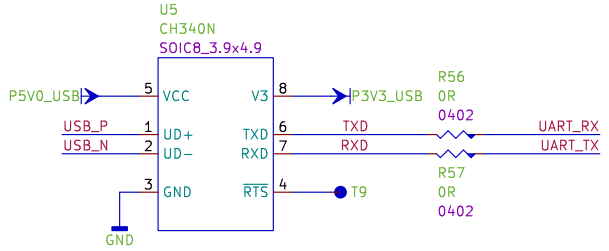
Page Title: **HDMI**

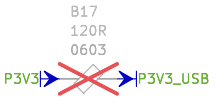
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File: hdmi.kicad_sch			Page: 9/12

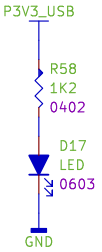


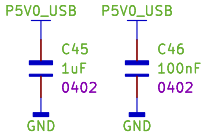


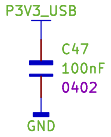




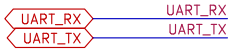












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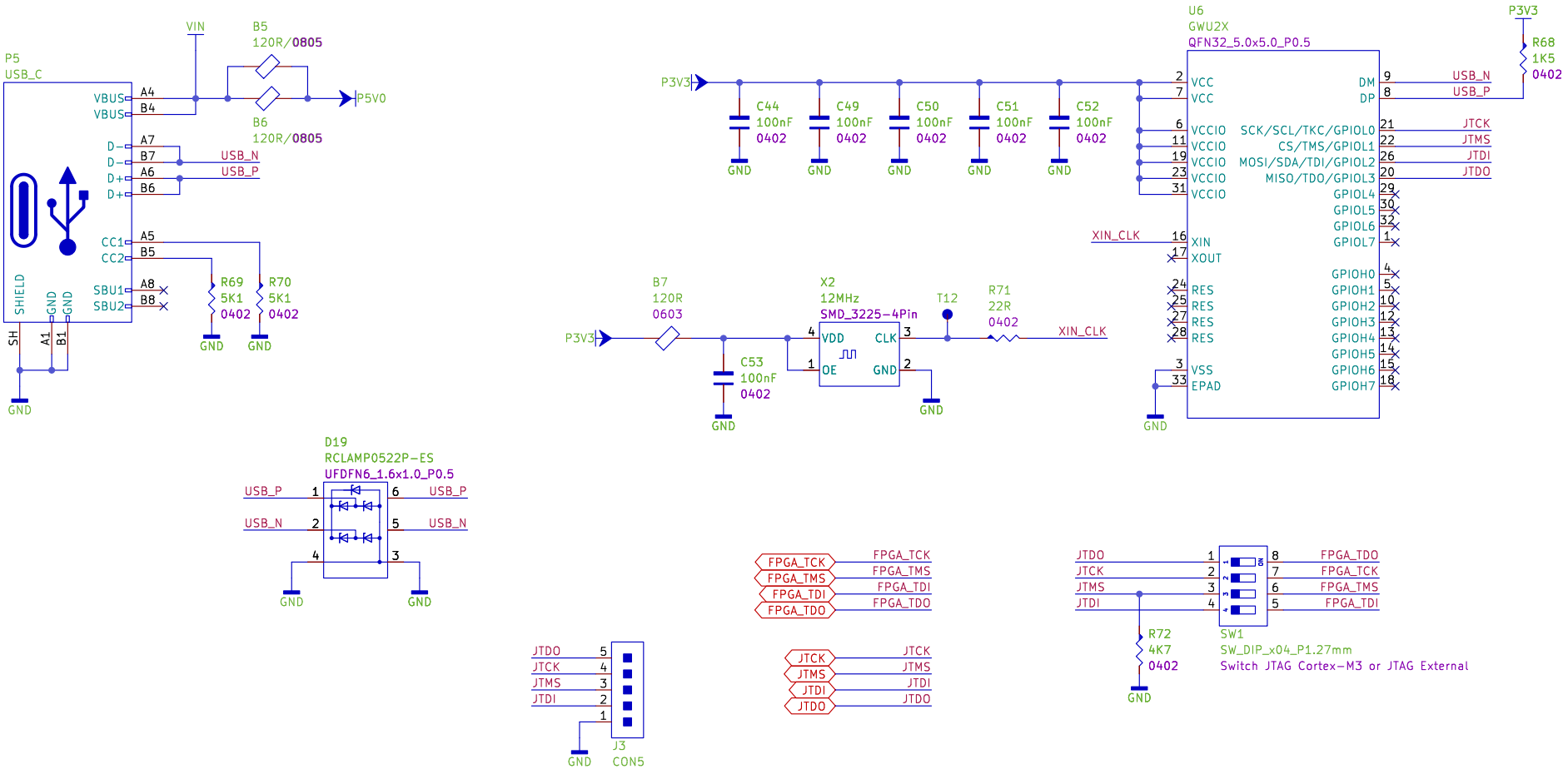
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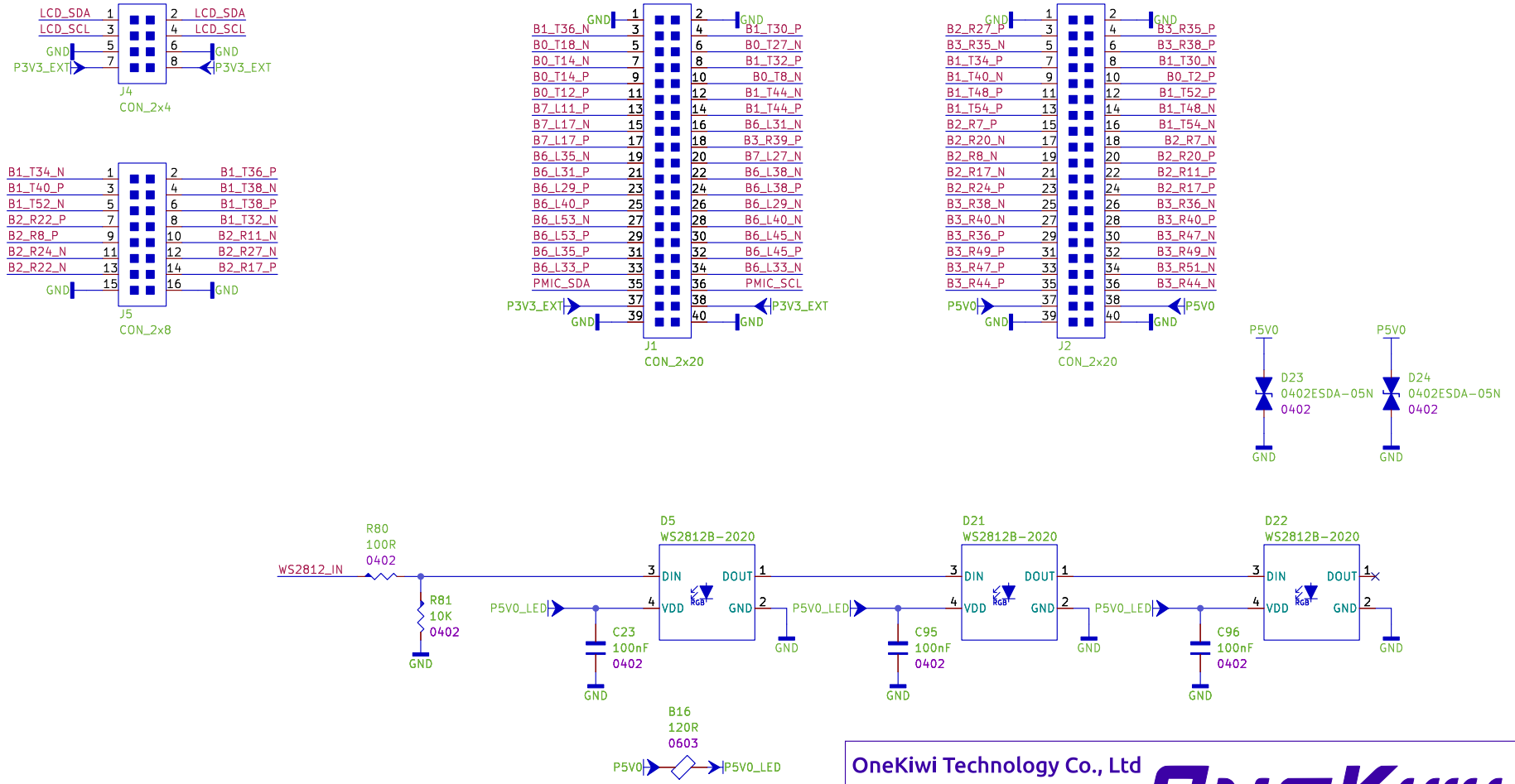
Drawing Title: **Kiwi Primer 20K**

Page Title: **UART**

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Drawing Title: ***Kiwi Primer 20K***			
Page Title: **JTAG**			
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Drawing Title: ***Kiwi Primer 20K***

Page Title: **Pinout**

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File: pinout.kicad_sch			Page: 12/12